

Lecture 3: Polynomial Interpolation

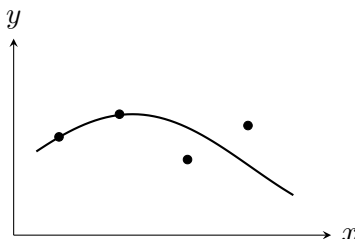
CSE 330 – Numerical Methods (Class Notes)

1 Polynomial Interpolation

We often describe relationships using *equations*, e.g.

$$x^2 + 2x + 2 = 0, \quad x^3 + 2x^2 + 5 = 0.$$

In interpolation we instead want a polynomial that passes through a given set of data points.



General polynomial of degree n :

$$P_n(x) = a_0x^0 + a_1x^1 + a_2x^2 + a_3x^3 + \cdots + a_nx^n.$$

Example ($n = 2$):

$$P_2(x) = a_0x^0 + a_1x^1 + a_2x^2$$

- Degree = 2.
- Coefficients = a_0, a_1, a_2 .
- Basis = $\{x^0, x^1, x^2\} = \{1, x, x^2\}$, which gives $n + 1$ functions that *span the space*.
- Dimension = 3 = # of basis functions = $(n + 1)$.

Function space and truncation error

A true function may need infinitely many terms, while a polynomial only keeps finitely many:

$$f(x) = \underbrace{(5 + 10x + 15x^2 + 20x^3)}_{\text{kept by } P_3(x)} + \underbrace{(25x^4 + \cdots \rightarrow \infty)}_{\text{missing = truncation error}}$$
$$P_3(x) = 5 + 10x + 15x^2 + 20x^3 \quad (\text{has a limitation}).$$

Here ∞ is the dimension / number of basis functions of the true function, so

$$f(x) \in V^\infty, \quad P_n(x) \in V^{n+1}.$$

Truncation error:

$$\text{Truncation error} = |f(x) - P_n(x)|.$$

As $n \uparrow$, the error reduces.

Weierstrass Approximation Theorem: as the polynomial order n increases, the approximation error reduces.

2 Taylor Series

$$f(x) = f(x_0) + f'(x_0)(x - x_0) + \frac{f''(x_0)(x - x_0)^2}{2!} + \frac{f'''(x_0)(x - x_0)^3}{3!} + \dots$$

Proof

Assume f can be written as a power series about x_0 :

$$f(x) = a_0 + a_1(x - x_0) + a_2(x - x_0)^2 + a_3(x - x_0)^3 + \dots$$

Differentiate repeatedly (a constant differentiates to 0):

$$\frac{d}{dx} : f'(x) = a_1 + 2a_2(x - x_0) + 3a_3(x - x_0)^2 + \dots$$

$$\frac{d}{dx} : f''(x) = 2a_2 + 3 \cdot 2a_3(x - x_0) + \dots$$

$$f'''(x) = 3 \cdot 2a_3 + \dots$$

Side note (how the power rule works):

$$\frac{d}{dx}(5x^2 + 2) = 5 \cdot 2x^1 + 0.$$

Now evaluate at $x = x_0$:

$$f(x_0) = a_0,$$

$$f'(x_0) = a_1,$$

$$f''(x_0) = 2a_2 \quad \implies \quad a_2 = \frac{f''(x_0)}{2!},$$

$$f'''(x_0) = 3 \cdot 2a_3 \quad \implies \quad a_3 = \frac{f'''(x_0)}{3!},$$

$$a_4 = \frac{f^{(4)}(x_0)}{4!}.$$

Putting $a_0, a_1, a_2, \dots$ back into $f(x)$, we get the Taylor series.

3 Example: Taylor Series of $\sin(x)$

Take $f(x) = \sin(x)$ with $x_0 = 0$. The Taylor series is

$$f(x) = f(x_0) + f'(x_0)(x - x_0) + \frac{f''(x_0)(x - x_0)^2}{2} + \frac{f'''(x_0)(x - x_0)^3}{3!} + \dots$$

With $x_0 = 0$ and using $\sin(0) = 0$, $\cos(0) = 1$, $-\sin(0) = 0$, $-\cos(0) = -1$:

$$\begin{aligned}\sin(x) &= \underbrace{\sin(0)}_0 + \underbrace{\cos(0)}_1(x - 0) + \underbrace{(-\sin(0))}_0 \frac{(x - 0)^2}{2!} + \underbrace{(-\cos(0))}_{-1} \frac{(x - 0)^3}{3!} + \dots \\ &= 0 + x - \frac{x^3}{3!} + \dots\end{aligned}$$

General form with error bound

Splitting into an n -degree polynomial plus a remainder (the start of the *error bound*):

$$f(x) = \sum_{k=0}^n \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k + \underbrace{\frac{f^{(n+1)}(\xi)}{(n+1)!} (x - x_0)^{n+1}}_{\text{T.S. error (remainder)}}$$

where $\xi \in [0, 0.5]$.